

Launch and Deployment Sequence

The interval between separation from the launch deployer and the start of routine operations is the highest-risk phase of a CubeSat's life. Every event in this window is scripted in advance, since the spacecraft is tumbling, thermally uncontrolled, and has no ground contact until its antenna deploys and its first beacon is heard. This document lays out a typical event sequence, the early power profile operators watch most closely, and the deployment milestone table used to track progress against plan.

Deployment Event Sequence

Immediately after separation, a preprogrammed inhibit delay holds off any deployments or radio transmissions for a short period, giving the spacecraft time to clear the deployer before anything extends. From there, the sequence below runs autonomously; no ground command is required until after first ground acquisition.

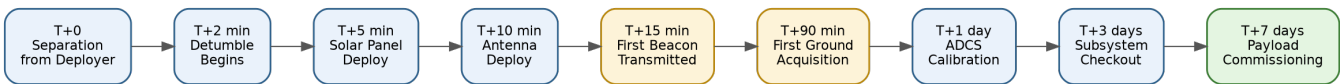


Figure 1. Deployment event timeline from separation through payload commissioning.

Deployment Milestone Table

Event	Time Post-Separation	Primary Subsystem
Separation from deployer	T+0	Structure
Detumble begins	T+2 min	ADCS
Solar panel deploy	T+5 min	Power
Antenna deploy	T+10 min	Communications
First beacon transmitted	T+15 min	Communications / OBC
First ground acquisition	T+90 min	Communications / Ground Segment
ADCS calibration complete	T+1 day	ADCS
Subsystem checkout complete	T+3 days	All subsystems
Payload commissioning begins	T+7 days	Payload

Early Power Profile

Battery state of charge is the single most closely watched telemetry parameter during the first few orbits, since the spacecraft is not yet in a controlled sun-pointing attitude and panel illumination is still irregular while detumble and ADCS calibration proceed. Each eclipse period draws the battery down; each sunlit arc recovers it. Operators expect a gradual net rise in the state-of-charge baseline across the first several orbits as attitude control improves and panel pointing becomes more consistent — a battery that fails to recover between eclipses during this window is treated as an early commissioning anomaly and investigated immediately rather than waiting for the standard fault-counter threshold to trip.

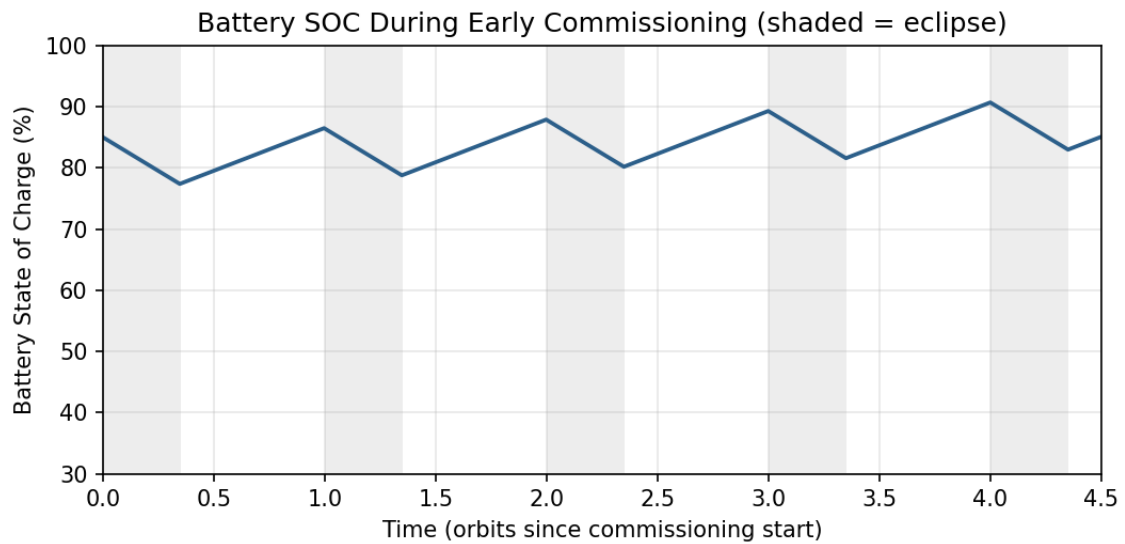


Figure 2. Battery state of charge across the first commissioning orbits, with eclipse periods shaded.

This early power telemetry, together with the deployment milestone table, forms the baseline that later anomaly detection compares against: once the spacecraft settles into nominal operations, any parameter drifting away from the pattern established here is what flags a developing issue, as described in the telemetry and anomaly detection and mission case study documents.